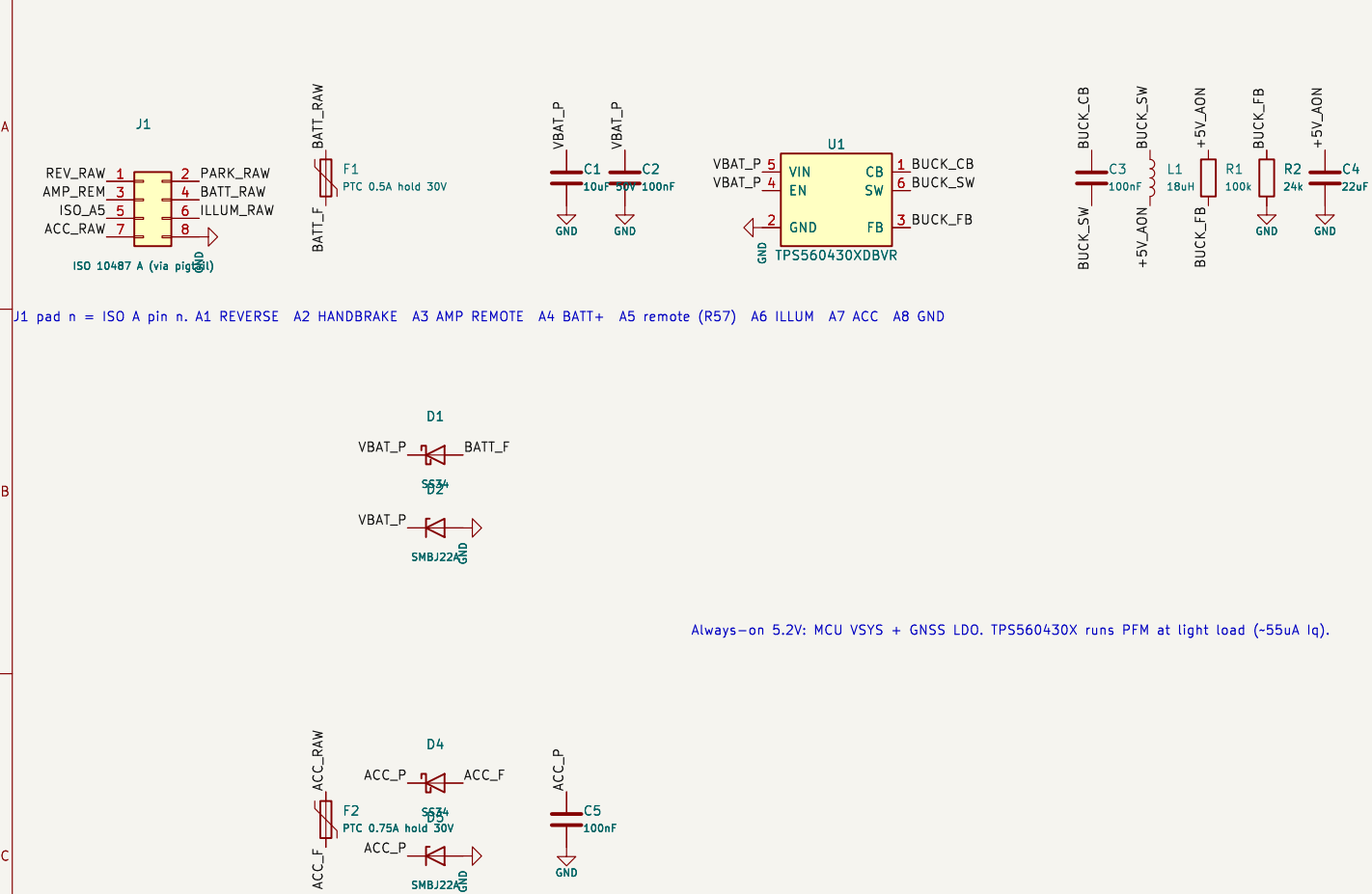
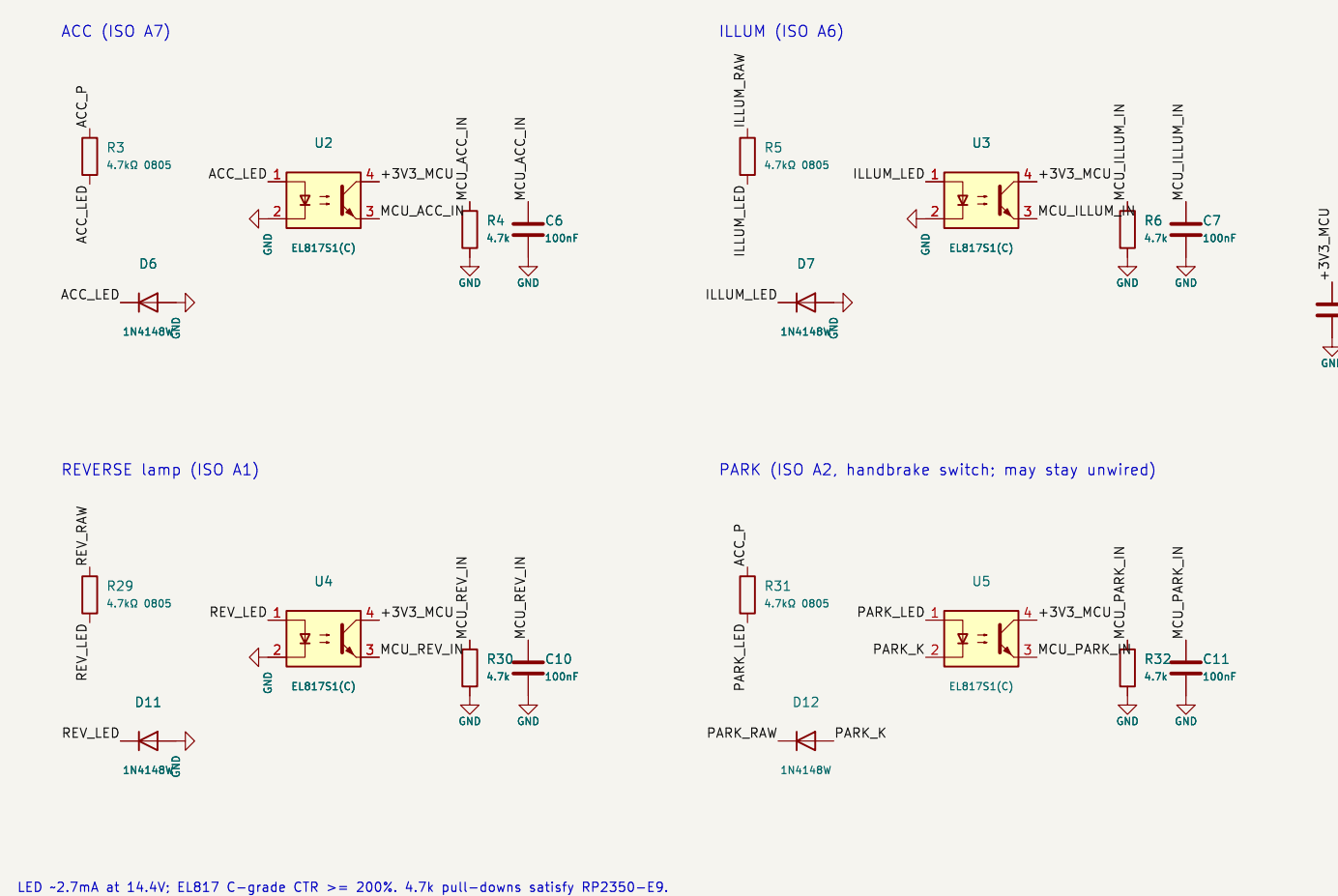


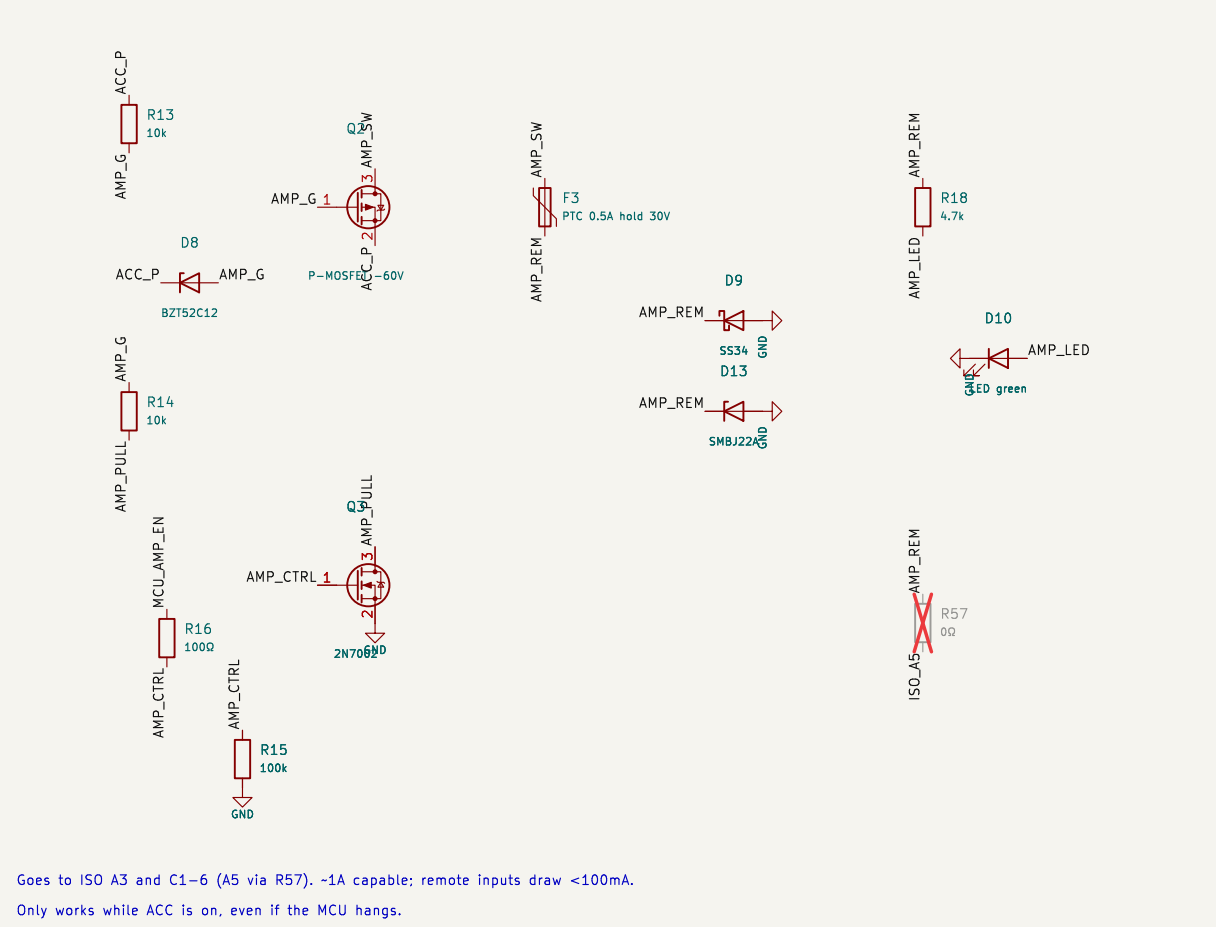
1. ISO 10487 CONNECTOR A (power) + ALWAYS-ON 5V FOR PICO AND GNSS



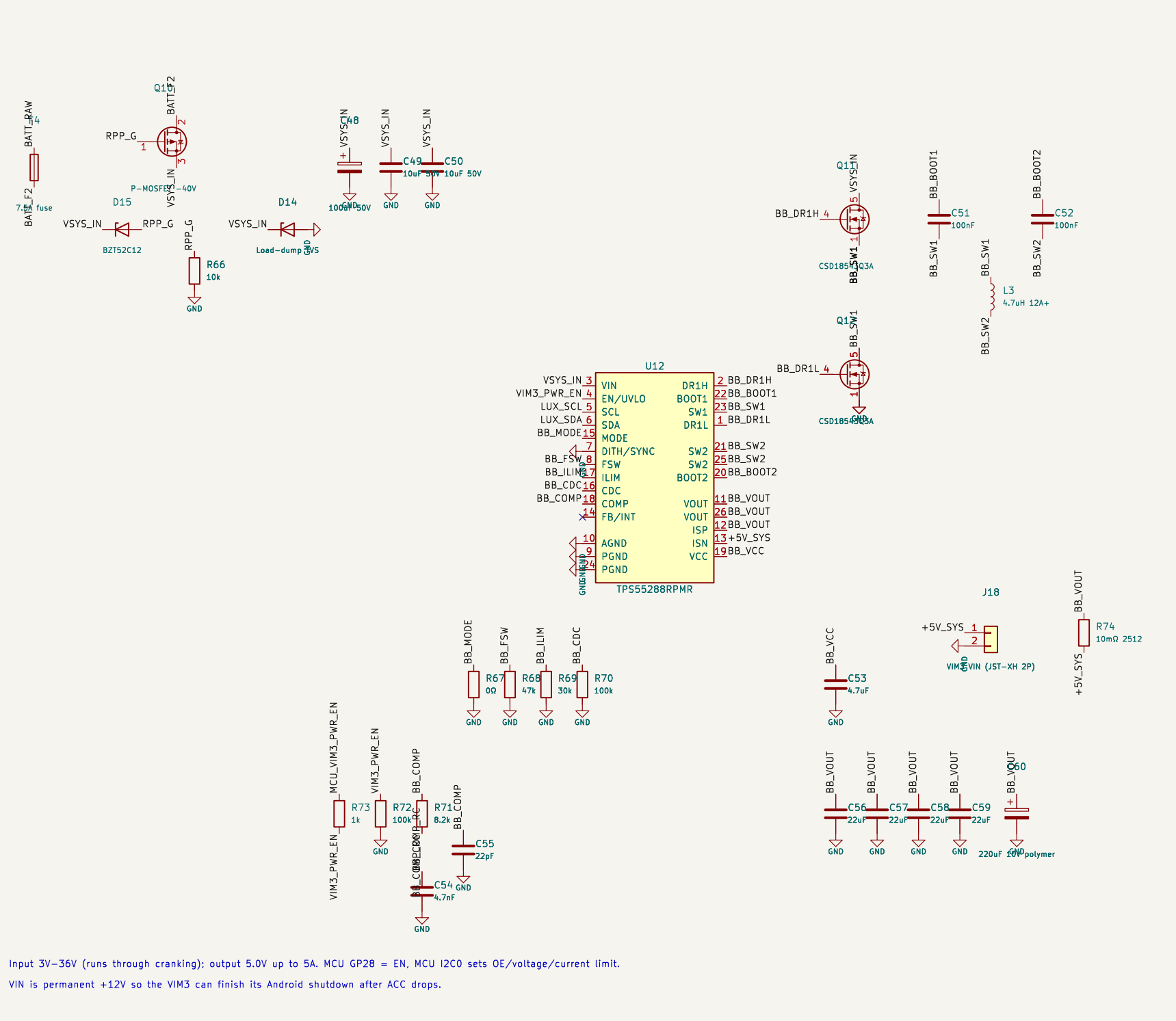
2. VEHICLE INPUTS (opto-isolated). ACC, ILLUM, REVERSE: +12V = on. PARK: switch to ground = on.



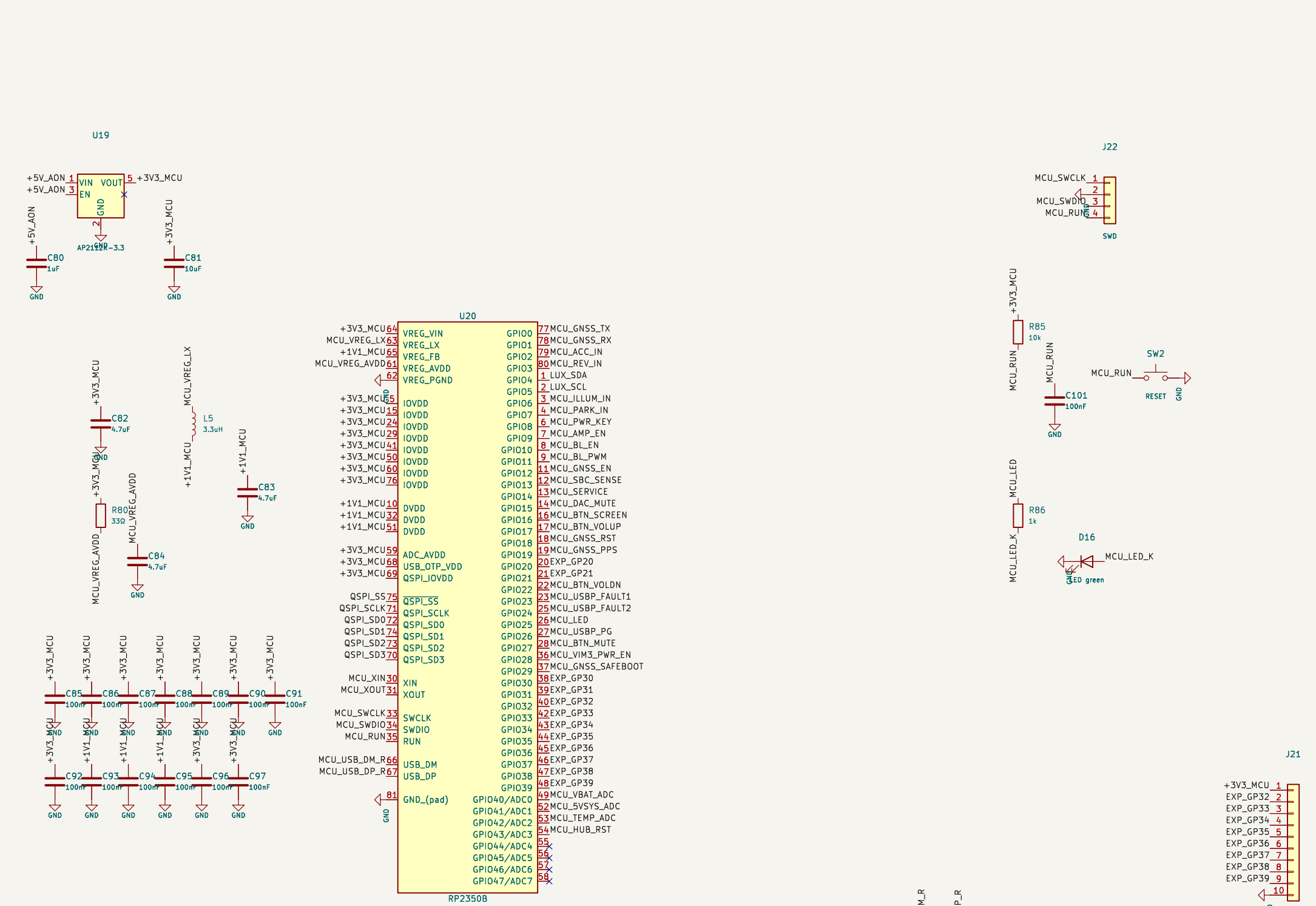
3. AMPLIFIER REMOTE (+12V from ACC, MCU GP9)



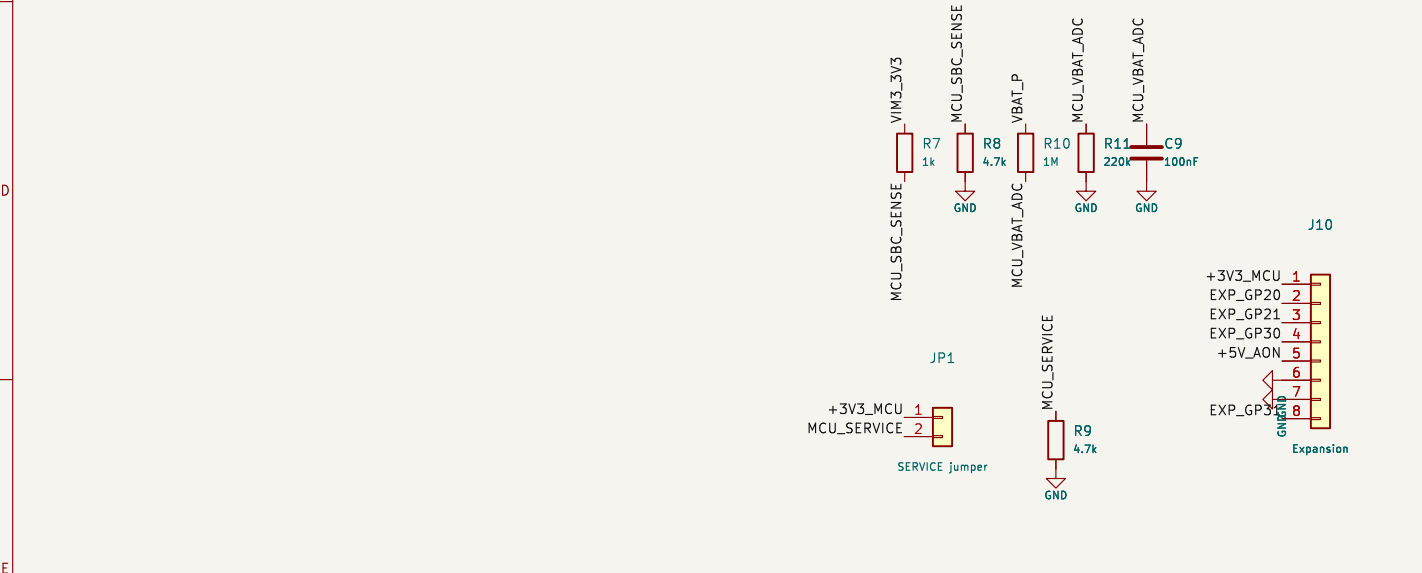
10. 5V/5A SYSTEM SUPPLY FOR VIM3 + SCREEN + HUB (TPS55288 buck-boost, rides through cranking and overvoltage)



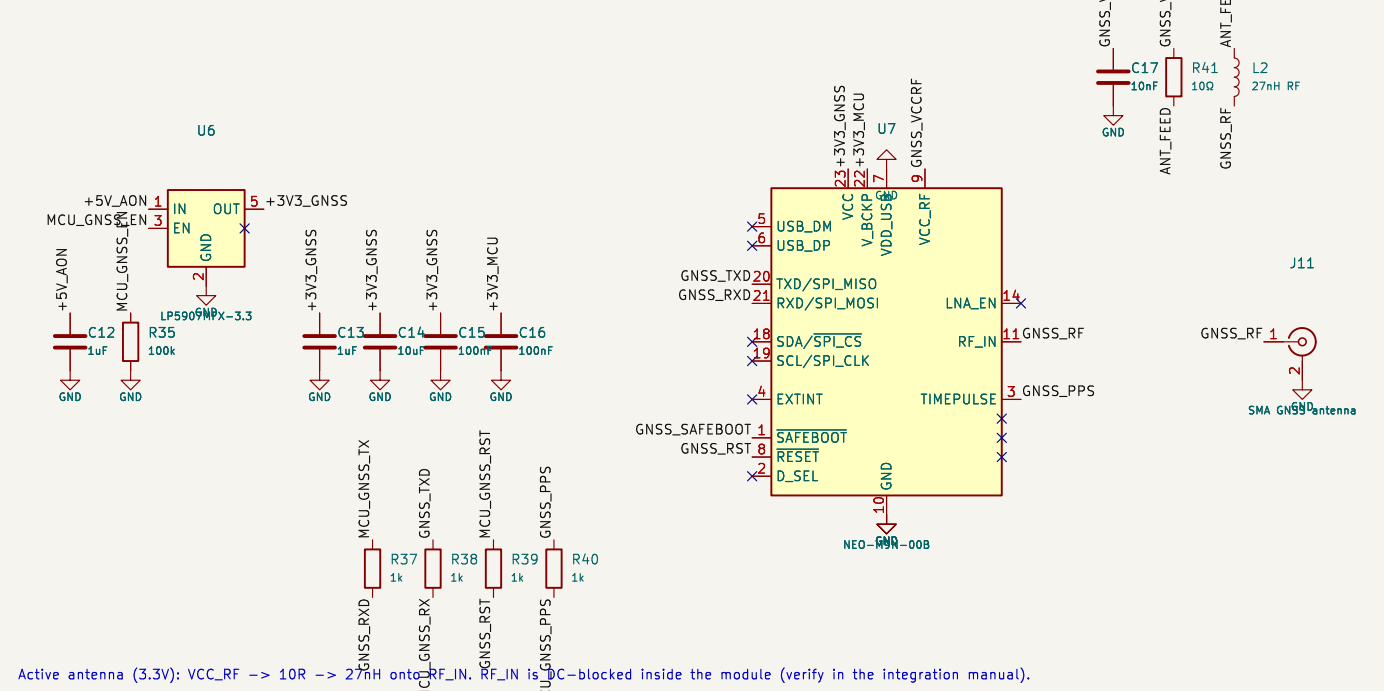
13. MCU: RP2350B (QFN-80) + 16MB QSPI flash + 12MHz crystal + 3.3V LDO, per the Raspberry Pi Pico 2 reference design



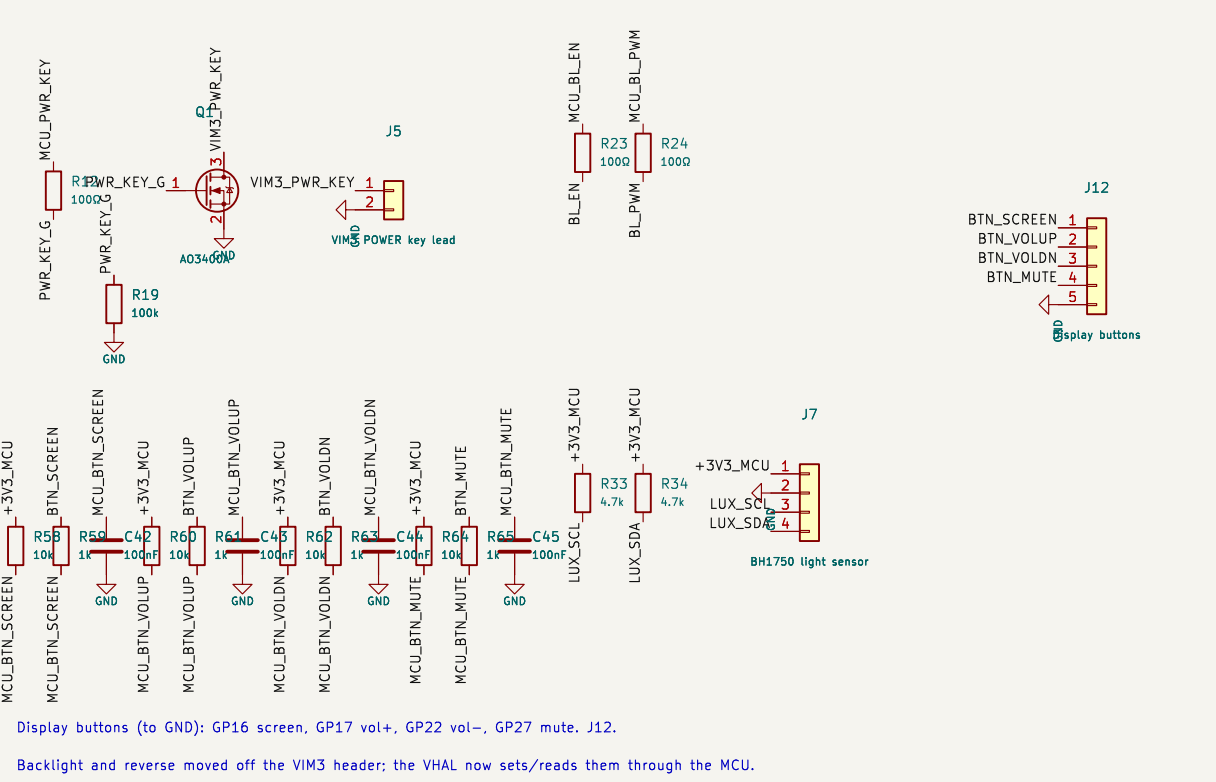
4. MCU HELPERS (MCU core is section 13)



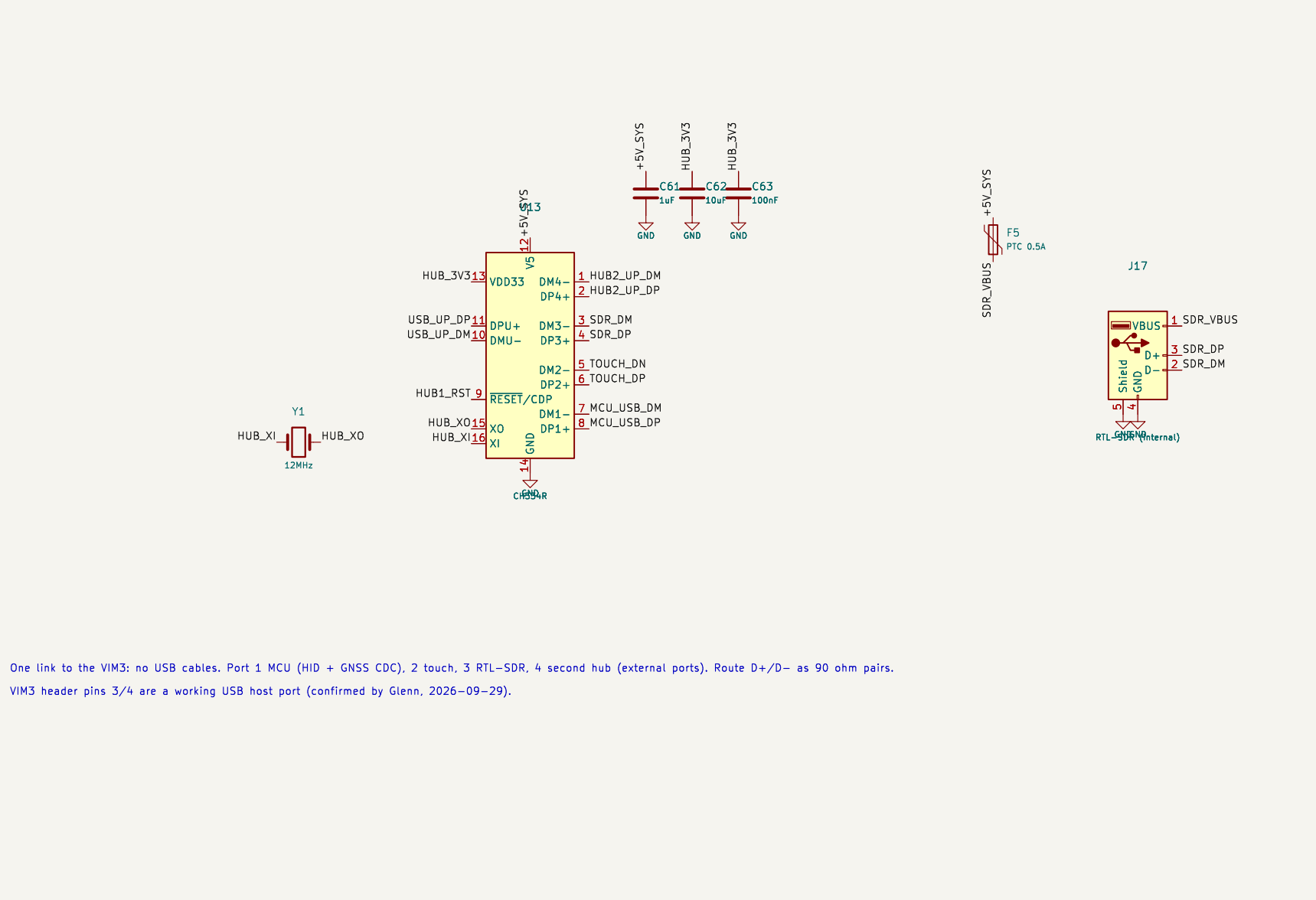
5. GNSS: u-blox NEO-M9N, active antenna on SMA



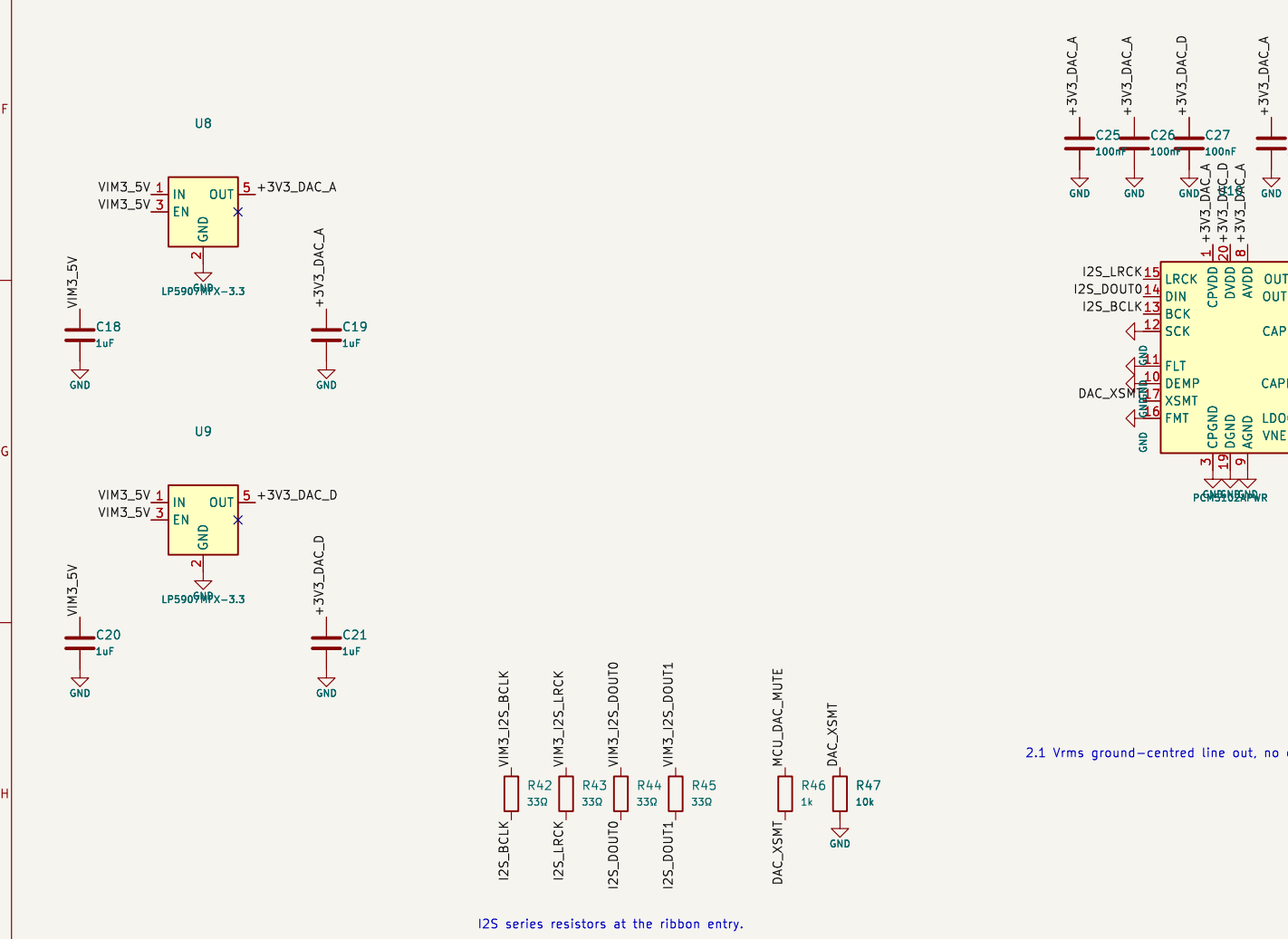
6. VIM3 POWER KEY, BACKLIGHT, LIGHT SENSOR (all from the MCU)



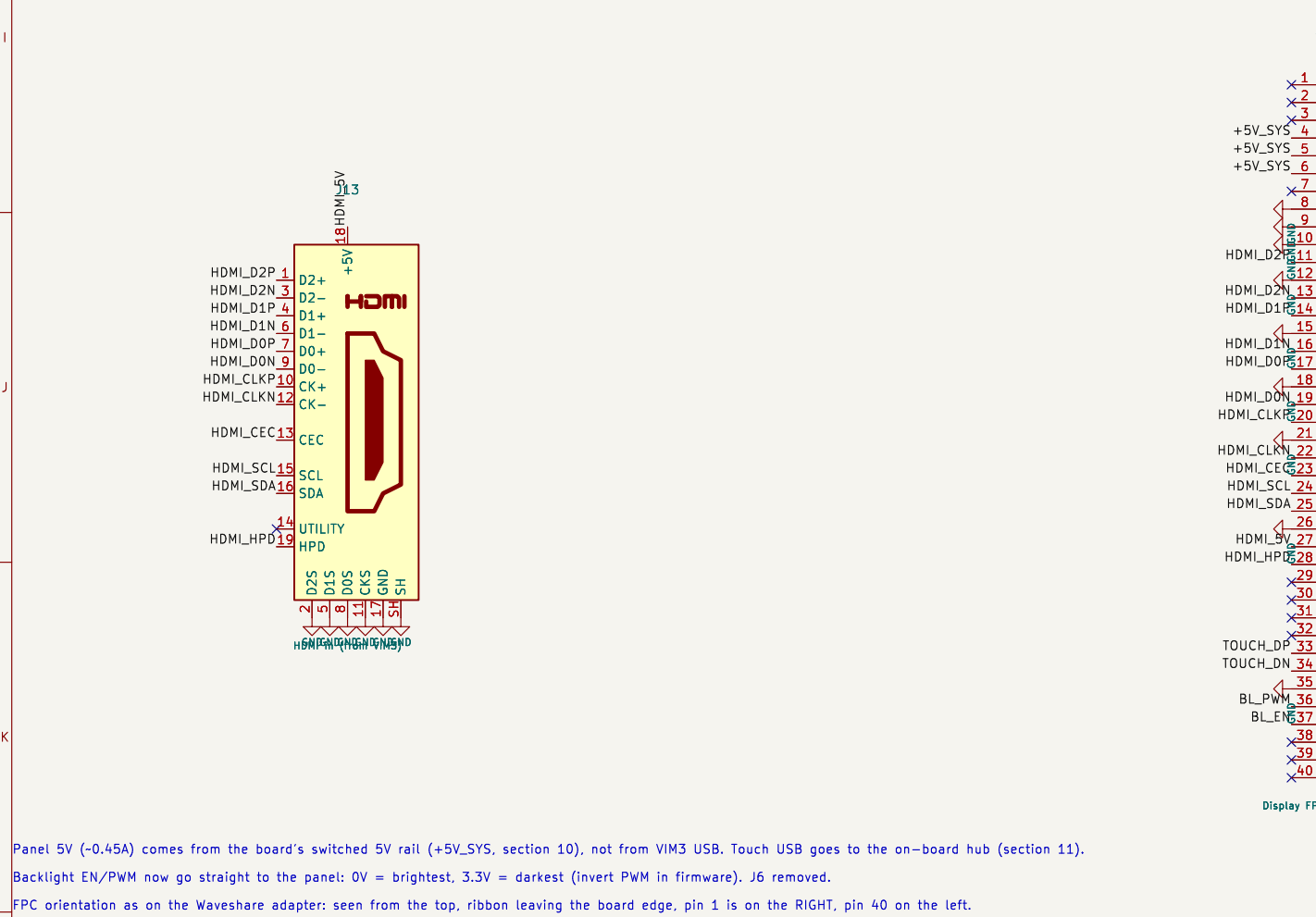
11. USB HUB: VIM3 header USB (pins 3/4) → MCU, screen touch, spare port



7. AUDIO: VIM3 TDM-B I2S -> 2x PCN5102A (front + rear) -> line out on ISO C1



9. DISPLAY: Waveshare 7in 70H-1024600 adapter built in (HDMI in from VM3 -> 40-pin FPC to the panel)



12. EXTERNAL USB PORTS (phone etc.): second hub + dedicated 5V/3A supply, 1.5A current-limited per port

